

### THRUST ROD AND LUG PLATE SCHEDULE

| PIPE DIA (mm) | TEST PRESSURE (Kpa)                                          | THRUST RODS (NO.-DIA) | RING FLANGE CLASS | RING FLANGE THICKNESS | RING FLANGE BOLTS |     |         | LUG PLATE |    |
|---------------|--------------------------------------------------------------|-----------------------|-------------------|-----------------------|-------------------|-----|---------|-----------|----|
|               |                                                              |                       |                   |                       | NO.               | DIA | MATL    | TL        | TE |
| 150           | 1.5 TIMES PRESSURE RATING ACCORDING TO THE RING FLANGE CLASS | 2-16                  | BS EN 1092 PN 25  | STANDARD              | 8                 | 20  | A307    | 12        | 29 |
|               |                                                              | 2-16                  |                   | STANDARD              | 8                 | 20  | A193-B7 | 12        | 29 |
|               |                                                              | 2-16                  |                   | STANDARD              | 8                 | 20  | A193-B7 | 16        | 29 |
|               |                                                              | 6-16                  | ANSI CL 300       | STANDARD              | 12                | 20  | A193-B7 | 16        | 25 |
| 200           |                                                              | 2-16                  | BS EN 1092 PN 25  | STANDARD              | 8                 | 20  | A307    | 12        | 32 |
|               |                                                              | 2-16                  |                   | STANDARD              | 8                 | 20  | A193-B7 | 16        | 32 |
|               |                                                              | 4-16                  |                   | STANDARD              | 8                 | 20  | A193-B7 | 16        | 32 |
|               |                                                              | 6-16                  | ANSI CL 300       | STANDARD              | 12                | 22  | A193-B7 | 20        | 29 |
| 250           |                                                              | 2-16                  | BS EN 1092 PN 25  | STANDARD              | 12                | 22  | A307    | 16        | 35 |
|               |                                                              | 3-16                  |                   | STANDARD              | 12                | 22  | A193-B7 | 16        | 35 |
|               |                                                              | 6-16                  |                   | STANDARD              | 12                | 22  | A307    | 16        | 35 |
|               |                                                              | 8-20                  | ANSI CL 300       | STANDARD              | 16                | 25  | A307    | 20        | 35 |
| 300           |                                                              | 2-22                  | BS EN 1092 PN 25  | STANDARD              | 12                | 22  | A193-B7 | 20        | 29 |
|               |                                                              | 3-22                  |                   | STANDARD              | 12                | 22  | A193-B7 | 20        | 29 |
|               |                                                              | 6-20                  |                   | STANDARD              | 12                | 22  | A193-B7 | 20        | 29 |
|               |                                                              | 8-20                  | ANSI CL 400       | STANDARD              | 16                | 32  | A307    | 20        | 41 |
| 350           |                                                              | 2-22                  | BS EN 1092 PN 25  | STANDARD              | 12                | 25  | A193-B7 | 20        | 35 |
|               |                                                              | 4-20                  |                   | STANDARD              | 12                | 25  | A307    | 20        | 35 |
|               |                                                              | 10-16                 | ANSI CL 300       | STANDARD              | 20                | 29  | A307    | 16        | 38 |
|               |                                                              | 10-22                 | ANSI CL 400       | STANDARD              | 20                | 32  | A307    | 20        | 41 |

#### NOTES:

- THRUST RODS SHALL CONFORM TO ASTM A307. NUTS SHALL CONFORM TO ASTM A563.
- LUG PLATES SHALL CONFORM TO ASTM A36.
- THRUST ROD NUTS SHALL BE TIGHTENED GRADUALLY AND EQUALLY IN STAGES TO PREVENT UNEVEN ALIGNMENT AND TO ALLOW EQUAL STRESS ON ALL RODS UNDER PRESSURE. TIGHTEN UNTIL SNUG. THREADS SHALL PROTRUDE FROM NUTS. PEEN THREADS AFTER TIGHTENING NUTS.
- RING FLANGES SHALL CONFORM TO BS EN 1092 PN 25 RATING CLASS 300, 400 OR 600 AS INDICATED. NOTE THAT FLANGE THICKNESS IS THICKER THAN STANDARD FOR 30, 36 AND 42 INCH PIPE FLANGES OPERATING AT 225 AND 375 PSI.
- LUG PLATES SHALL BE EQUALLY SPACED AROUND PIPE.
- PROVIDE CATHODIC PROTECTION FOR COUPLINGS AS NOTED ON PLANS OR IN THE SPECIFICATIONS.

### THRUST TIE FOR STEEL PIPE - SCHEDULE

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STANDARD DETAIL

### THRUST ROD AND LUG PLATE SCHEDULE

| PIPE DIA (mm) | TEST PRESSURE (Kpa)                                          | THRUST RODS (NO.-DIA) | RING FLANGE CLASS | RING FLANGE THICKNESS | RING FLANGE BOLTS |     |         | LUG PLATE |    |
|---------------|--------------------------------------------------------------|-----------------------|-------------------|-----------------------|-------------------|-----|---------|-----------|----|
|               |                                                              |                       |                   |                       | NO.               | DIA | MATL    | TL        | TE |
| 400           | 1.5 TIMES PRESSURE RATING ACCORDING TO THE RING FLANGE CLASS | 4-20                  | BS EN 1092 PN 25  | STANDARD              | 16                | 25  | A307    | 20        | 35 |
|               |                                                              | 8-16                  |                   | STANDARD              | 16                | 25  | A307    | 16        | 35 |
|               |                                                              | 8-20                  |                   | STANDARD              | 16                | 25  | A193-B7 | 20        | 35 |
|               |                                                              | 10 - 25               | ANSI CL 600       | STANDARD              | 20                | 38  | A307    | 22        | 50 |
| 450           |                                                              | 4-20                  | BS EN 1092 PN 25  | STANDARD              | 16                | 29  | A307    | 20        | 38 |
|               |                                                              | 8-20                  |                   | STANDARD              | 16                | 29  | A307    | 16        | 38 |
|               |                                                              | 8-22                  |                   | STANDARD              | 16                | 29  | A307    | 20        | 38 |
|               |                                                              | 10-25                 | ANSI CL 600       | STANDARD              | 20                | 41  | A307    | 22        | 54 |
| 600           |                                                              | 4-25                  | BS EN 1092 PN 25  | STANDARD              | 20                | 32  | A307    | 22        | 41 |
|               |                                                              | 10-20                 |                   | STANDARD              | 20                | 32  | A307    | 20        | 41 |
|               |                                                              | 10-25                 |                   | STANDARD              | 20                | 32  | A307    | 22        | 41 |
|               |                                                              | 12-30                 | ANSI CL 600       | STANDARD              | 24                | 48  | A307    | 25        | 60 |
| 750           |                                                              | 7-25                  | BS EN 1092 PN 25  | STANDARD              | 28                | 32  | A307    | 22        | 41 |
|               |                                                              | 14-22                 |                   | STANDARD              | 28                | 32  | A307    | 20        | 41 |
|               |                                                              | 14-27                 |                   | STANDARD              | 28                | 44  | A307    | 22        | 57 |
|               |                                                              | 14-35                 |                   | STANDARD              | 28                | 44  | A307    | 29        | 57 |
| 900           |                                                              | 8-29                  | BS EN 1092 PN 25  | STANDARD              | 32                | 38  | A307    | 22        | 50 |
|               |                                                              | 16-22                 |                   | STANDARD              | 32                | 38  | A307    | 20        | 50 |
|               |                                                              | 16-30                 |                   | STANDARD              | 32                | 50  | A307    | 22        | 67 |
|               |                                                              | 16-38                 |                   | STANDARD              | 32                | 50  | A307    | 29        | 67 |
| 1050          |                                                              | 12-25                 | BS EN 1092 PN 25  | STANDARD              | 36                | 38  | A307    | 22        | 50 |
|               |                                                              | 18-25                 |                   | STANDARD              | 36                | 38  | A307    | 22        | 50 |
|               |                                                              | 18-32                 |                   | STANDARD              | 36                | 50  | A307    | 25        | 67 |

- CONTRACTOR SHALL USE SCHEDULE DATA FOR ONLY THOSE PIPE SIZES AND TEST PRESSURES SPECIFIED IN THIS CONTRACT.
- FLANGE BOLTS SHALL CONFORM TO ASTM A307 WITH NUTS PER A563 OR ASTM A193 GRADE B7 WITH NUTS PER A194 GRADE 7 AS INDICATED.
- WHERE ALTERNATIVE MATERIALS OF CONSTRUCTION ARE USED FOR MORE CORROSSIVE OR AGGRESSIVE ENVIRONMENTS, THE TENSILE STRENGTH OF THESE ALTERNATIVE MATERIALS SHALL NOT BE INFERIOR TO ASTM A307 OR ASTM

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